

RUSHI ASHOK SAVANI

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SUMMARY

ML engineer specializing in production AI systems for healthcare, with experience spanning predictive modeling, large language model workflows, and cloud-native deployment. Proven ability to take models from development through validation to production release in regulated, compliance-sensitive environments. Seeking full-lifecycle ownership of ML infrastructure where engineering rigor directly drives operational and financial outcomes.

WORK EXPERIENCE

AI/ML Engineering Intern | *ProHealth Group, Atlanta GA*

Jun 2025 – May 2026

- Architected a production RAG system with Pinecone vector search, Claude-driven SQL generation, and per-user query caching that serves clinical and financial insights to healthcare administrators, enabling natural language data lookup under 10 seconds.
- Trained an XGBoost rehospitalization classifier on 12 clinical and claims features with SMOTE-Tomek resampling to correct a 4:1 class imbalance, lifting risk-targeting precision from 64% to 82% across a 55K patient holdout set.
- Shipped LLM-powered investigation workflows inside the launched Careflux Analytics SaaS product with citations for source traceability, iterative prompt engineering and structured eval plans to validate anomaly detection outputs, saving an estimated \$100K/year in manual analysis.
- Provisioned AWS infrastructure through Terraform, containerized services in multi-stage Docker builds, and orchestrated rolling deploys via EC2 Auto Scaling Groups with GitHub Actions CI/CD and CloudWatch observability.

Research Assistant | *Syracuse University, School of Information Studies*

Jan 2025 – May 2025

- Fine-tuned a RoBERTa-base model through HuggingFace Transformers on 5000+ labeled AI-adoption survey responses, segmenting five user clusters with 91% classification accuracy to quantify organizational readiness gaps.
- Automated data cleaning and statistical analysis in pandas and scipy, piping transformed outputs into Power BI dashboards that cut the research team's report cycle from two weeks to three days.

Data Analyst | *Krishna Jewels, India*

Jun 2023 - Jun 2024

- Designed a relational data model and ontology across 15+ MySQL tables and 40K+ records across inventory, sales, suppliers, and accounting records, digitizing paper-based tracking and reducing lookup time by 70%.
- Surfaced slow-moving SKUs and high-demand items through SQL-based sales trend analysis, improving gross margin by 12%.
- Centralized transactional data into Power BI dashboards with daily scheduled refreshes, giving stakeholders self-serve access and eliminating 15+ hours/week of manual reporting.

TECHNICAL SKILLS

- Programming Languages:** Python, SQL, React.js, Typescript
- Machine Learning & Statistics:** Scikit-learn, XGBoost, PyTorch, TensorFlow, Feature Engineering, Model Evaluation, Hyperparameter Tuning, Statistical Modeling, A/B Testing.
- GenAI & NLP:** LLMs, RAG, LangChain, LangGraph, Vector Embeddings (Pinecone, Qdrant), LiteLLM, Vertex AI/Gemini, HuggingFace Transformer, SpaCy, Text Classification, Sentiment Analysis (RoBERTa).
- Cloud & MLOps:** AWS (EC2, ECS, S3, Secrets Manager), Terraform, Docker, GitHub Actions CI/CD, FastAPI, REST APIs
- Data Engineering & BI:** Data Modeling, DBT, Fivetran, ETL Pipelines, PostgreSQL, MySQL, MS SQL, Power BI, Tableau.

EDUCATION

Master of Science | Information Systems, Data Science

Syracuse University, School of Information Studies, Syracuse, NY

May 2026

(GPA: 3.9/4)

Bachelor of Engineering | Information Technology

Thadomal Shahani Engineering College, India

May 2024

(GPA: 8.52/10)

PROJECTS

VaultMind | *Private On-premises Q&A System*

- Deployed a zero-egress RAG system for document Q&A on quantized Llama 3.1 with nomic-embed-text embeddings and Qdrant vector search, serving fully offline inference through FastAPI to meet healthcare data privacy requirements.
- Optimized retrieval accuracy for long clinical documents by layering BM25 sparse matching with dense vector search via Reciprocal Rank Fusion and reranking top candidates through a cross-encoder, cutting p50 latency by 68% on a 3B quantized local model and eliminating ~\$140/month in cloud LLM costs.

MyFinancial Advisor | *Investment Recommendation System*

- Forecasted short-term stock direction using an LSTM network on engineered price and market-signal features, outperforming a logistic-regression baseline by 14% in directional accuracy.
- Indexed 500+ financial filings and news articles as vector embeddings in Pinecone, enabling source-backed validation of investment signals through a RAG retrieval layer.

KarmaScore Predictor | *Subscription Churn Analysis*

- Predicted subscriber churn across 10K+ records using XGBoost, reducing 60 raw features to 23 through importance ranking and correlation filtering to achieve 89% precision on the churn class.
- Mapped SHAP scores into a customer-level intervention likelihood metric ranking which at-risk subscribers are worth retaining based on projected profitability, displayed in an interactive Streamlit dashboard.

Annual Report Analysis with Advanced Language Models: A Stock Investment Strategy Enhancement

- Processed data from 50+ organizational annual reports and deployed OpenAI's ChatGPT-3.5 Turbo for generating structured responses to financial analyst queries on investment trends and report interpretation and published findings in (IRJET)